Instrument Closure and Multipartite Selection:
Exact Counterexamples and Conditional Completion Theorems

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Abstract

A two-system convex family can reproduce ordinary elementary complex two-level quantum statistics while restricting composite preparations. An earlier four-system counterexample showed that a uniform convex interpolation between fully separable and all density operators fails parallel-preparation closure. We strengthen this in four ways. First, every nonendpoint interpolation already fails closure under a *single-party successful local filter*, and the violation and its nonzero success probability are computed exactly. Second, permitting a different mixture parameter at four systems cannot restore *both* parallel preparation and partial trace. Third, an explicit matching-generated hierarchy provides a positive countercontrol: it is closed under parallel preparation, discarding and elementary local trace-preserving operations, retains entangled two-system preparations and may violate CHSH, but excludes a three-system GHZ state. Fourth, within a stipulated complex density-operator representation, closure under local successful filters and local unitaries makes *one entangled pure two-system seed* sufficient to fill the complete two-system state space. A related mixed-state extension invokes published distillation results and substantially stronger assumptions. Every statement is conditional mathematical research, not an emergence of operational probability, physical systems, time, quantum mechanics, classical mechanics, or a calibrated prephysical-to-physical map.

1 Assessment contract and definitions

The program's target is an explained emergence of quantum and classical mechanics from independently warranted prephysical structure, with an empirical map. Here the *mathematical control* is assumed: an elementary Hilbert space $H = \mathbb{C}^2$ with its usual complex inner product, positive density operators, Born-form evaluation, Hilbert tensor products and the standard meaning of a quantum operation. These are *stipulations of the test*, not derived primitive phenomena and not identification of physical qubits. Let $\mathcal{D}_n$ be the density operators on $H^{\otimes n}$, and

$$\mathsf{Sep}_n = \operatorname{conv}\{\rho_1 \otimes \cdots \otimes \rho_n : \rho_j \in \mathcal{D}_1\}$$

the fully separable subset. For $0 \le t \le 1$, define the Minkowski mixture

$$\mathcal{K}_t^{(2)} = (1-t)\mathsf{Sep}_2 + t\mathcal{D}_2 = \{(1-t)\sigma + t\tau : \sigma \in \mathsf{Sep}_2,\ \tau \in \mathcal{D}_2\}. \tag{1}$$

It is closed under convex mixing and elementary local completely positive trace-preserving (CPTP) operations. Its extreme $t = 0$ is a *separable composite with quantum elementary systems*, not classical Hamiltonian mechanics. Its $t = 1$ endpoint is the ordinary quantum two-system composite. Set

$$|\Phi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad P = |\Phi\rangle\langle\Phi|, \qquad \rho_t = (1-t)|00\rangle\langle 00| + tP.$$

We use $0 \preceq E \preceq \mathbf{1}$ as positive operator inequalities. A linear separation functional need not already be implemented as a physical effect. When operational access is discussed, the needed admissible instrument is stated.

Lemma 1 (Sharp Bell support)**.** *For every* $0 \le t \le 1$*,*

$$\max_{\rho \in \mathcal{K}_t^{(2)}} \operatorname{Tr}(P\rho) = \frac{1+t}{2}.$$

The state ρ_t *attains the maximum.*

Proof. For product unit vectors $|a\rangle|b\rangle$, Cauchy–Schwarz gives $|\langle\Phi|a\otimes b\rangle|^2 \le 1/2$; convexity extends the bound to Sep_2. The maximum over $\mathcal{D}_2$ is 1. The affine mixture has upper bound $(1-t)/2 + t$, attained by the separable $|00\rangle\langle 00|$ and the quantum P constituents. □

2 Instrument-level obstruction at two systems

Theorem 2 (A single successful local filter leaves every interior family)**.** *For every* $0 < t < 1$*,* $\mathcal{K}_t^{(2)}$ *is not closed under normalized nonzero successful branches of all single-party local quantum instruments. In fact, an ordinary two-outcome instrument on the first subsystem has a branch with success probability exactly* t*, whose conditioned output is outside* $\mathcal{K}_t^{(2)}$*.*

Proof. For $0 < x \le 1$ use Kraus operators on the first system

$$F_x = \begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}, \qquad F_x^{\mathrm{fail}} = \begin{pmatrix} \sqrt{1-x^2} & 0 \\ 0 & 0 \end{pmatrix}, \qquad F_x^\dagger F_x + (F_x^{\mathrm{fail}})^\dagger F_x^{\mathrm{fail}} = \mathbf{1}_2.$$

Applying the successful branch $F_x \otimes \mathbf{1}_2$ to ρ_t gives the unnormalized state

$$\widetilde{\rho}_{t,x} = (1-t)x^2|00\rangle\langle 00| + \frac{t}{2}(x|00\rangle + |11\rangle)(x\langle 00| + \langle 11|),$$

with success probability

$$p_{t,x} = \operatorname{Tr}\widetilde{\rho}_{t,x} = \frac{(2-t)x^2 + t}{2} > 0. \tag{2}$$

The conditioned Bell-projector expectation is

$$f_{t,x} = \frac{\operatorname{Tr}(P\widetilde{\rho}_{t,x})}{p_{t,x}} = \frac{1}{2} + \frac{tx}{(2-t)x^2 + t}. \tag{3}$$

The derivative vanishes at $x_* = \sqrt{t/(2-t)} \in (0,1)$, where it is maximal on $(0,1]$. At this point

$$p_{t,x_*} = t, \qquad f_{t,x_*} = \frac{1+\sqrt{t/(2-t)}}{2}, \qquad f_{t,x_*} - \max_{\rho\in\mathcal{K}_t^{(2)}} \operatorname{Tr}(P\rho) = \frac{\sqrt{t/(2-t)} - t}{2} > 0. \tag{4}$$

Strict positivity follows from $\frac{t}{2-t} - t^2 = \frac{t(1-t)^2}{2-t} > 0$. Lemma 1 therefore excludes the normalized output from $\mathcal{K}_t^{(2)}$. This is a violation of *postselected-state closure*, not of trace-preserving channel closure. □

Remark 3 (Access, limits and controls)**.** *At* $t = 1/2$*,* $x_* = 1/\sqrt{3}$*, success probability is* $1/2$*,* $f_{t,x_*} = (1+1/\sqrt{3})/2$*, and the gap is* $0.03867513459\ldots$*. As* $t \downarrow 0$ *success probability tends to zero; the endpoint* $t = 0$ *is not assigned a nonexistent normalized zero-probability branch. Both* $\mathcal{K}_0^{(2)} = \mathsf{Sep}_2$ *and* $\mathcal{K}_1^{(2)} = \mathcal{D}_2$ *are closed under nonzero-probability local filters. Near* $t \uparrow 1$ *the positive gap is* $\frac{1}{4}(1-t)^2 + O((1-t)^3)$*; floating-point zero is not a mathematical counterexample. The Bell witness is accessible by* product-observable statistics *if local Pauli measurements are supplied, since* $P = \frac{1}{4}(\mathbf{1}\otimes\mathbf{1} + X\otimes X - Y\otimes Y + Z\otimes Z)$*. Those effects, outcome probabilities, postselection, Pauli measurements and preparation resources are all imported in this mathematical test.*

3 An arity-dependent interpolation still fails at four systems

Allow two and four elementary systems to use *different* mixture weights $t, s \in [0,1]$:

$$\mathcal{K}_t^{(2)} = (1-t)\mathsf{Sep}_2 + t\mathcal{D}_2, \qquad \mathcal{K}_s^{(4)} = (1-s)\mathsf{Sep}_4 + s\mathcal{D}_4.$$

Theorem 4 (Independent preparation and discarding are incompatible)**.** *If (i) the tensor product of any two allowed pair states is admitted at four systems and (ii) discarding either disjoint pair maps every four-system state into $\mathcal{K}_t^{(2)}$, then necessarily $t = s \in \{0,1\}$. In particular no interior pair weight can be repaired by choosing an independent four-body weight s of the displayed form.*

Proof. Discard bound. For any s, the state $\rho_s^{12} \otimes |00\rangle\langle 00|^{34}$ belongs to $\mathcal{K}_s^{(4)}$: its fully separable and unrestricted density constituents can be chosen with the same spectator $|00\rangle^{34}$. Discarding systems 3,4 gives ρ_s^{12}; Lemma 1 implies

$$s \le t. \tag{5}$$

Parallel-preparation bound. Let $P_{12} = P \otimes \mathbf{1}_4$ and $P_{34} = \mathbf{1}_4 \otimes P$ and define the projector

$$E = \mathbf{1}_{16} - (\mathbf{1}_4 - P) \otimes (\mathbf{1}_4 - P).$$

For any fully product pure four-party state, the two disjoint pair Bell probabilities f_1, f_2 satisfy $f_i \le 1/2$. Thus $\mathrm{Tr}(E\omega) = 1 - (1-f_1)(1-f_2) \le 3/4$ for all $\omega \in \mathsf{Sep}_4$. The upper bound is attained by $|0000\rangle$; $\max_{\mathcal{D}_4} \mathrm{Tr}\, E = 1$. Consequently

$$\max_{\omega \in \mathcal{K}_s^{(4)}} \mathrm{Tr}(E\omega) = \frac{3}{4} + \frac{s}{4}.$$

Independent preparation of ρ_t^{12} and ρ_t^{34} requires $\rho_t \otimes \rho_t \in \mathcal{K}_s^{(4)}$, whose actual witness value is $1 - ((1-t)/2)^2$. Hence

$$s \ge 2t - t^2. \tag{6}$$

Together (5) and (6) require $t \ge 2t - t^2$, i.e. $t(1-t) \le 0$. With $t \in [0,1]$, this allows only $t = 0$ or 1, and the two inequalities then force $s = t$. □

Remark 5. *The theorem excludes precisely a* uniform-at-each-arity Minkowski interpolation *with weights allowed to vary by arity. It does not exclude all possible four-system extensions. The witness E can serve as an ordinary linear separation functional even if a proposed instrument theory does not physically implement it. For the special case $s = t$, the violation is $t(1-t)/4$, recovering the earlier four-party counterexample. The new contradiction uses an additional* discarding *obligation and defeats attempts to change s to repair that example.*

4 Positive countercontrol: a coherent restricted hierarchy

A no-go must not be promoted beyond its hypotheses. Let $\mathfrak{P}_n$ be the finite set of partitions of $\{1, \dots, n\}$ into blocks of size one or two. Set $L_1 = \mathcal{D}_1$, $L_2 = \mathcal{K}_t^{(2)}$ and define

$$\mathcal{M}_n(t) = \mathrm{conv} \bigcup_{\pi \in \mathfrak{P}_n} \left\{ \bigotimes_{b \in \pi} \rho_b : \rho_b \in L_{|b|} \right\}. \tag{7}$$

The tensor factors are placed on their labeled subsystems. This is a *matching-generated* family, not the interpolation of Theorem 4.

Theorem 6 (Consistency under specified weak operations)**.** *For every $t \in [0,1]$, the hierarchy* (7) *is compact, convex, invariant under permutations and closed under independent preparation, discarding any subset of elementary systems, and arbitrary* unconditioned *one-site CPTP operations. It also permits destructive single-site measurements with postselection* if the measured site is removed. *Its single-system and two-system spaces are $\mathcal{D}_1$ and $\mathcal{K}_t^{(2)}$. Product-observable tomography distinguishes its density operators. Nevertheless $\mathcal{M}_3(t) \ne \mathcal{D}_3$ for every t.*

Proof. Each block-state set is compact, and there are finitely many partitions; finite-dimensional convex hull preserves compactness. Permutations relabel partitions. The tensor product of two block-product terms is a block-product term on the disjoint union; bilinear expansion extends this to convex mixtures. Partial traces either remove whole blocks or turn a surviving member of a two-block into an arbitrary $\mathcal{D}_1$ state. On-site CPTP operations preserve L_1 and preserve L_2 because they separately preserve the fully separable and quantum constituents. A destructive single-site effect similarly leaves its block partner in $\mathcal{D}_1$ and other blocks unchanged, with normalized mixtures remaining admissible. Product Pauli effects span the full Hermitian operator space and hence separate any two different states.

Let $|\mathrm{GHZ}\rangle = (|000\rangle + |111\rangle)/\sqrt{2}$ and $Q = |\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|$. Every block-product term at three systems factors across some one-site versus two-site bipartition. Across every such split the Schmidt coefficients of $|\mathrm{GHZ}\rangle$ are $(1/\sqrt{2}, 1/\sqrt{2})$. Thus $\mathrm{Tr}(Q\omega) \leq 1/2$ for every $\omega \in \mathcal{M}_3(t)$, by the maximal Schmidt-overlap bound and convexity. But $\mathrm{Tr}(QQ) = 1$, so $Q \notin \mathcal{M}_3(t)$. □

Remark 7 (Not a full operational emergence theory)**.** *The hierarchy deliberately does* not *claim closure under nondestructive successful local filters, arbitrary joint instruments, entangling gates or joint operations on all members of one laboratory. Adding these assumptions requires a new closure proof and can defeat* (7)*. For every $t > 0$ it nevertheless contains the entangled ρ_t (its partial transpose has eigenvalue $-t/2$). For the stipulated Pauli CHSH settings it contains a state with $\langle\mathrm{CHSH}\rangle = \sqrt{2}(1+t) > 2$ whenever $t > \sqrt{2} - 1$. Thus all-arity parallel preparation, discard, local state tomography, and a Bell-CHSH violation do not alone force the unrestricted many-body density spaces. This is a restricted mathematical quantum subtheory with imported probabilities and operations,* not *a proposed physical ontology.*

5 A conditional positive completion mechanism

Theorem 8 (One entangled pure seed with local selective filters)**.** *Let $\mathcal{L} \subseteq \mathcal{D}_2$ be convex and invariant under all local unitaries and every normalized nonzero successful branch of a one-party local quantum filter $F \otimes \mathbf{1}_2$, where $F^\dagger F \preceq \mathbf{1}_2$. If $\mathcal{L}$ contains even one* pure entangled *two-system state, then $\mathcal{L} = \mathcal{D}_2$.*

Proof. By local unitaries write the seed in Schmidt form $|\psi\rangle = a|00\rangle + b|11\rangle$, with $a, b > 0$ and $a^2 + b^2 = 1$. Every target two-system unit vector has Schmidt form $(U \otimes V)(c|00\rangle + d|11\rangle)$ with $c, d \geq 0$, $c^2 + d^2 = 1$. Choose any $\kappa > 0$ small enough that $\kappa c/a \leq 1$ and $\kappa d/b \leq 1$. Then

$$F = \mathrm{diag}(\kappa c/a, \kappa d/b) \quad \text{satisfies} \quad F^\dagger F \preceq \mathbf{1}_2,$$

and $(F \otimes \mathbf{1})|\psi\rangle = \kappa(c|00\rangle + d|11\rangle)$. The branch has positive success probability κ^2; normalization, followed by the local $U \otimes V$, produces the target pure state. Thus $\mathcal{L}$ contains every pure bipartite state. Their convex hull is $\mathcal{D}_2$, while $\mathcal{L} \subseteq \mathcal{D}_2$ by hypothesis. □

Corollary 9 (Mixed entangled seed under stronger published premises)**.** *Suppose $\mathcal{L} \subseteq \mathcal{D}_2$ is also compact and is the two-system sector of an all-arity theory closed under tensor powers, discarding, normalized successful finite-round* LOCC *protocols on arbitrarily many copies, local unitaries and successful local filters. If $\mathcal{L}$ contains an entangled two-system density operator, then $\mathcal{L} = \mathcal{D}_2$,* conditional on the published theorem *that every entangled 2×2 quantum state is distillable by such protocols.*

Proof. The two-qubit distillation theorem of Horodecki, Horodecki and Horodecki [2] gives finite-copy LOCC outputs with Bell-state fidelity tending to one. The assumed operational closure places every normalized output in $\mathcal{L}$. Compactness places the limiting Bell projector in $\mathcal{L}$. Apply Theorem 8. □

Remark 10 (Assumption debt)**.** *Theorem 8 assumes a standard complex composite, a pure entangled seed, arbitrarily chosen local filters, successful-branch updating, and convex mixing. Corollary 9 adds all-arity systems, cross-copy laboratory operations, classical communication, postselection, a compactness limit, and an external distillation theorem. These assumptions can do the mathematical selection work but do not explain why a prephysical substrate supplies them. The statement is* not *that any CHSH violation, bare tensor closure, or holonomy alone implies full quantum theory.*

6 Reconciliation with the broader research objective

The projective holonomy determinant identity, its root-of-unity ambiguity, global conjugation equivalence, finite exact-homogeneity obstruction, and finite-cycle complex-type algebra remain conditional or restricted results of earlier work. Neither they nor the present composition theorems derive an operational clock, the Born rule, a Hamiltonian, physical spacetime, a quantum-to-classical Hamiltonian limit, or empirical prediction with uncertainty. The finite-cycle construction also admits a classical simplex completion; present composite results import complex Hilbert spaces and therefore *do not bridge that selection gap*.

The resulting obligation hierarchy is sharper: choose independently warranted primitive operations and records; identify probability and measurement update; justify composition and conditioning at every arity; identify any actual entangling preparation and instrument; then separately derive dynamics, physical units and a controlled classical limit. A failure of one natural interpolation does not exclude every completion; Theorem 6 supplies an explicit countercontrol. Work remains at Initial External Grounding. No ontology is selected.

Reproducibility and provenance

The accompanying source checker uses exact SymPy reductions, high-precision MPMath boundary cases, ordinary finite-dimensional matrix controls, a seeded batch of GHZ biseparable controls and sampled pure-state local filters. Theorems, rather than finite random tests, supply the universal quantifiers. The one-party local-filter result and arity-varying four-party bound have exact algebraic proofs; the sampling checks do not estimate a physical distribution. All boundary and numerical outlier cases are retained.

Research direction and scholarly responsibility: Kirtus Justus. Mathematical development, code execution and internal checking assisted by ChatGPT (GPT-6). This is one-assistant internal work and incorporates other assistant-generated submissions as *internal research provenance*, not as independent refereeing. The underlying research release was not modified; no experimental results, independent scientific validation or host-enforced research protocol are claimed.

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